

Lab04

Rafael Nóbrega Costa - RA:296730

Lab04

1.

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
```

2.

```
fname <- "C:/Users/aftran/Documents/ME315/Lab04/WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

3.

```
countries_raw <- read_excel(fname, sheet = "Country")
```

4.

```
countries <- countries_raw |> filter(`Country Code` %in% c("BRA", "ABW", "AFE", "AFG", "AFW"))
countries
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name`      `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>          <chr>        <chr>      <chr>
1 ABW           Aruba           Aruba        Aruba      AW
2 AFE           Africa Eastern and Sou~ Africa East~ Africa Eas~ ZH
3 AFG           Afghanistan      Afghanistan Islamic St~ AF
4 AFW           Africa Western and Cen~ Africa West~ Africa Wes~ ZI
5 BRA           Brazil           Brazil       Federative~ BR
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

5.

```
WDI_raw <- read_excel(fname, sheet = "Data")
```

6.

```
WDI <- WDI_raw |> select(`Country Code`, `Indicator Code`, `1960`:`2018`)
WDI
```

```
# A tibble: 396,970 x 61
  `Country Code` `Indicator Code` `1960` `1961` `1962` `1963` `1964` `1965`
  <chr>          <chr>          <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
1 AFE           EG.CFT.ACCS.ZS      NA     NA     NA     NA     NA     NA
2 AFE           EG.CFT.ACCS.RU.ZS   NA     NA     NA     NA     NA     NA
3 AFE           EG.CFT.ACCS.UR.ZS   NA     NA     NA     NA     NA     NA
4 AFE           EG.ELC.ACCS.ZS      NA     NA     NA     NA     NA     NA
5 AFE           EG.ELC.ACCS.RU.ZS   NA     NA     NA     NA     NA     NA
6 AFE           EG.ELC.ACCS.UR.ZS   NA     NA     NA     NA     NA     NA
```

```

7 AFE          FX.OWN.TOTL.ZS          NA      NA      NA      NA      NA      NA
8 AFE          FX.OWN.TOTL.FE.ZS        NA      NA      NA      NA      NA      NA
9 AFE          FX.OWN.TOTL.MA.ZS        NA      NA      NA      NA      NA      NA
10 AFE          FX.OWN.TOTL.OL.ZS        NA      NA      NA      NA      NA      NA
# i 396,960 more rows
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, ...

```

7.

```

WDI <- WDI |> pivot_longer(cols = "1960":"2018",
                           names_to = "Year",
                           names_transform = list(Year = as.integer)) |> pivot_wider(names_from = "Year",
                           values_from = "value")
WDI

```

```

# A tibble: 15,635 x 1,500
  `Country Code` Year EG.CFT.ACCS.ZS EG.CFT.ACCS.RU.ZS EG.CFT.ACCS.UR.ZS
  <chr>          <int>      <dbl>          <dbl>          <dbl>
1 AFE            1960          NA              NA              NA
2 AFE            1961          NA              NA              NA
3 AFE            1962          NA              NA              NA
4 AFE            1963          NA              NA              NA
5 AFE            1964          NA              NA              NA
6 AFE            1965          NA              NA              NA
7 AFE            1966          NA              NA              NA
8 AFE            1967          NA              NA              NA
9 AFE            1968          NA              NA              NA
10 AFE           1969          NA              NA              NA
# i 15,625 more rows
# i 1,495 more variables: EG.ELC.ACCS.ZS <dbl>, EG.ELC.ACCS.RU.ZS <dbl>,
#   EG.ELC.ACCS.UR.ZS <dbl>, FX.OWN.TOTL.ZS <dbl>, FX.OWN.TOTL.FE.ZS <dbl>,
#   FX.OWN.TOTL.MA.ZS <dbl>, FX.OWN.TOTL.OL.ZS <dbl>, FX.OWN.TOTL.40.ZS <dbl>,
#   FX.OWN.TOTL.PL.ZS <dbl>, FX.OWN.TOTL.60.ZS <dbl>, FX.OWN.TOTL.SO.ZS <dbl>,
#   FX.OWN.TOTL.YG.ZS <dbl>, per_si_allsi.adq_pop_tot <dbl>,
#   per_allsp.adq_pop_tot <dbl>, per_sa_allsa.adq_pop_tot <dbl>, ...

```

8.

```
select(WDI, Year)
```

```
# A tibble: 15,635 x 1
```

```
  Year
<int>
1  1960
2  1961
3  1962
4  1963
5  1964
6  1965
7  1966
8  1967
9  1968
10 1969
```

```
# i 15,625 more rows
```

9.

```
WDI <- WDI |> select(`Country Code`, Year, SP.DYN.CBRT.IN) |> filter(`Country Code` %in% countries)
Data <- left_join(countries, WDI)
```

Joining with `by = join\_by(`Country Code`)`

```
Data
```

```
# A tibble: 295 x 33
```

```
  `Country Code` `Short Name` `Table Name` `Long Name` `2-alpha code`
<chr>           <chr>        <chr>      <chr>      <chr>
1 ABW            Aruba        Aruba      Aruba      AW
2 ABW            Aruba        Aruba      Aruba      AW
3 ABW            Aruba        Aruba      Aruba      AW
4 ABW            Aruba        Aruba      Aruba      AW
5 ABW            Aruba        Aruba      Aruba      AW
6 ABW            Aruba        Aruba      Aruba      AW
7 ABW            Aruba        Aruba      Aruba      AW
8 ABW            Aruba        Aruba      Aruba      AW
9 ABW            Aruba        Aruba      Aruba      AW
10 ABW           Aruba        Aruba      Aruba      AW
```

```
# i 285 more rows
```

```
# i 28 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

10.

```
Data |> arrange(`Country Code`, Year)
```

```
# A tibble: 295 x 33
  `Country Code` `Short Name` `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>        <chr>      <chr>      <chr>
1 ABW           Aruba        Aruba      Aruba      AW
2 ABW           Aruba        Aruba      Aruba      AW
3 ABW           Aruba        Aruba      Aruba      AW
4 ABW           Aruba        Aruba      Aruba      AW
5 ABW           Aruba        Aruba      Aruba      AW
6 ABW           Aruba        Aruba      Aruba      AW
7 ABW           Aruba        Aruba      Aruba      AW
8 ABW           Aruba        Aruba      Aruba      AW
9 ABW           Aruba        Aruba      Aruba      AW
10 ABW          Aruba        Aruba      Aruba      AW
# i 285 more rows
# i 28 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

11.

```
ggplot(Data, aes(x = Year, y = SP.DYN.CBRT.IN, colour = `Country Code`)) +
  geom_line()
```

